

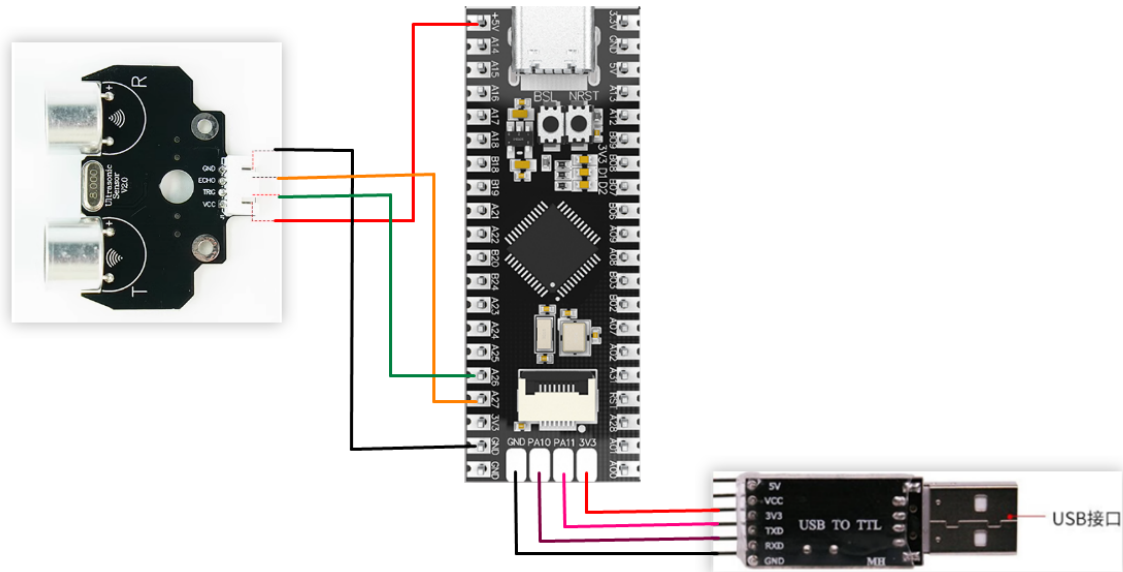
Ultrasonic Ranging

I. Learning Objectives

Measure distance using an ultrasonic module and output the data to a serial monitor on the computer via the serial port.

II. Hardware Connections

Wiring the Ultrasonic Module and USB-to-TTL Adapter to the MSPM0G3507



Note: If you do not have a TTL module, you can also use a USB-C serial connection directly.

III. Program Explanation

- uart.h

```
#ifndef __USART_H__
#define __USART_H__

#include "ALLHeader.h"

void USART_Init(void);

void uart0_send_char(char ch);
void uart0_send_string(char* str);

#endif
```

Header file defining serial transmission of characters and strings.

- uart.c

```

//串口发送单个字符
void uart0_send_char(char ch)
{
    //当串口0忙的时候等待，不忙的时候再发送传进来的字符
    while( DL_UART_isBusy(UART_0_INST) == true );
    //发送单个字符
    DL_UART_Main_transmitData(UART_0_INST, ch);
}

//串口发送字符串
void uart0_send_string(char* str)
{
    //当前字符串地址不在结尾 并且 字符串首地址不为空
    while(*str!=0&&str!=0)
    {
        //发送字符串首地址中的字符，并且在发送完成之后首地址自增
        uart0_send_char(*str++);
    }
}

//串口的中断服务函数
void UART_0_INST_IRQHandler(void)
{
    uint8_t receivedData = 0;

    //如果产生了串口中断
    switch( DL_UART_getPendingInterrupt(UART_0_INST) )
    {
        case DL_UART_IIDX_RX://如果是接收中断

            // 接收发送过来的数据保存
            receivedData = DL_UART_Main_receiveData(UART_0_INST);

            // 检查缓冲区是否已满
            if (recv0_length < RE_0_BUFF_LEN_MAX - 1)
            {
                recv0_buff[recv0_length++] = receivedData;

                // 将保存的数据再发送出去，不想回传可以注释掉
                uart0_send_char(receivedData);
            }
            else
            {
                recv0_length = 0;
            }

            // 标记接收标志
            recv0_flag = 1;

            break;

        default://其他的串口中断
            break;
    }
}

```

Sending a character and a string using USART0, and the serial port interrupt service routine.

- delay.h

```
#ifndef _DELAY_H
#define _DELAY_H

#include <stdint.h>
#include "../ti_msp_dl_config.h"

void delay_us(unsigned long __us);
void delay_ms(unsigned long ms);

#endif
```

Header file defining functions for milliseconds and microseconds

- delay.c

```
#include "delay.h"

volatile unsigned int delay_times = 0;

//搭配滴答定时器实现的精确us延时
void delay_us(unsigned long __us)
{
    uint32_t ticks;
    uint32_t told, tnow, tcnt = 38;

    // 计算需要的时钟数 = 延迟微秒数 * 每微秒的时钟数
    ticks = __us * (32000000 / 1000000);

    // 获取当前的SysTick值
    told = SysTick->VAL;

    while (1)
    {
        // 重复刷新获取当前的SysTick值
        tnow = SysTick->VAL;

        if (tnow != told)
        {
            if (tnow < told)
                tcnt += told - tnow;
            else
                tcnt += SysTick->LOAD - tnow + told;

            told = tnow;

            // 如果达到了需要的时钟数, 就退出循环
            if (tcnt >= ticks)
                break;
        }
    }
}

//搭配滴答定时器实现的精确ms延时
```

```

void delay_ms(unsigned long ms)
{
    delay_us( ms * 1000 );
}

void SysTick_Handler(void)
{
    if(delay_times != 0)
    {
        delay_times--;
    }
}

```

Implementing millisecond- and microsecond-level delays using timer counting.

- ultrasonic.h

```

#ifndef __ULTRASONIC_H_
#define __ULTRASONIC_H_

#include "AllHeader.h"

#define SR04_TRIG(x) ( x ? DL_GPIO_setPins(TRIG_PORT,TRIG_PIN_12_PIN) : \
DL_GPIO_clearPins(TRIG_PORT,TRIG_PIN_12_PIN) )
#define SR04_ECHO() ( ( ( DL_GPIO_readPins(ECHO_PORT,ECHO_PIN_13_PIN) & \
ECHO_PIN_13_PIN ) > 0 ) ? 1 : 0 )

void Ultrasonic_Init(void);
float Hcsr04GetLength(void );

#endif

```

Define header files for ultrasonic initialization and ranging.

- ultrasonic.c

```

float Hcsr04GetLength(void)
{
    /*测5次数据计算一次平均值*/
    volatile float length = 0;
    t = 0;
    volatile float sum = 0;
    volatile unsigned int i = 0;

    Close_Timer();

    while(i != 5)
    {
        SR04_TRIG(0); //trig拉低信号，发出低电平s
        delay_us(10); //持续时间超过5us

        SR04_TRIG(1); //trig拉高信号，发出高电平

        delay_us(15); //持续时间超过10us
    }
}

```

```

        SR04_TRIG(0); //trig拉低信号，发出低电平

        /*Echo发出信号 等待回响信号*/
        /*输入方波后，模块会自动发射8个40KHz的声波，与此同时回波引脚（echo）端的电
平会由0变为1；
        （此时应该启动定时器计时）；当超声波返回被模块接收到时，回波引 脚端的电平会由
1变为0；
        （此时应该停止定时器计数），定时器记下的这个时间即为
        超声波由发射到返回的总时长；*/

        while(SR04_ECHO() == 0); //echo等待回响

        Open_Timer(); //打开定时器

        i++;

        while(SR04_ECHO() > 0);

        Close_Timer(); // 关闭定时器

        delay_ms(100);

        t = (float)Get_TIMER_Count(); // 获取时间,分辨率为1us
        length = (float)t / 58.0f; // cm
        sum += length;

    }
    length = sum/5; //五次平均值
    distance = length;
    return length;
}

```

Using precise timer-based delays, five ultrasonic measurements are obtained, and their average value is calculated and returned.

- empty.c

```

#include "AllHeader.h"

int main(void)
{

    USART_Init();

    Ultrasonic_Init();

    printf("Start.....\r\n");

    while(1)
    {
        uint32_t value = (int)Hcsr04GetLength();

        printf((const char *)"Distance = %dCM\r\n", value);

        delay_ms(1000);
    }
}

```

```
}
```

Initialize the serial port and the ultrasonic sensor, and print the ultrasonic measurement data to the serial port every 1000 milliseconds.

Note: The project source code must be placed in the SDK path for compilation.

For example, the path : D:\TI\M0_SDK\mspm0_sdk_1_30_00_03\1.TB6612

> 新加卷 (D:) > TI > M0_SDK > mspm0_sdk_1_30_00_03				
名称	修改日期	类型	大小	
1.TB6612	2024/7/22 18:59	文件夹		
2.AT8236	2024/7/22 19:47	文件夹		
3.Enconder	2024/7/23 10:36	文件夹		
4.Servo	2024/7/23 11:13	文件夹		
docs	2024/7/23 10:33	文件夹		
examples	2024/7/23 10:34	文件夹		
kernel	2024/7/23 10:37	文件夹		
source	2024/7/23 10:33	文件夹		
tools	2024/7/23 10:33	文件夹		
imports.mak	2024/1/25 11:45	MAK 文件	2 KB	
known_issues_FAQ.html	2024/1/25 11:42	Microsoft Edge ...	67 KB	
license_msp0_sdk_1_30_00_03.txt	2024/1/25 11:42	文本文档	33 KB	
manifest_msp0_sdk_1_30_00_03.html	2024/1/25 11:42	Microsoft Edge ...	113 KB	
mspm0sdk_1_30_00_03.log	2024/7/23 10:42	文本文档	5,237 KB	
release_notes_msp0_sdk_1_30_00_0...	2024/1/25 11:42	Microsoft Edge ...	108 KB	
uninstall.dat	2024/7/23 10:39	DAT 文件	344 KB	
uninstall.exe	2024/7/23 10:39	应用程序	6,048 KB	

IV. Experimental Results

To program the MSPM0G3507, first connect the wiring according to the diagram. Close any other programs using the serial port, then open a serial terminal application on your computer, select the appropriate COM port, and set the baud rate to 9600. You will see the distance data displayed in the terminal, with the unit in centimeters (cm).